

Gi-Cheon Kang

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Lab: heilaboratory.github.io

CURRENT POSITION	Assistant Professor , Ajou University Department of Software and Computer Engineering Directing Human-Centered Embodied Intelligence (HEI) Lab	2026 –
EDUCATION	Seoul National University Ph.D. in Artificial Intelligence <i>Advisor:</i> Byoung-Tak Zhang <i>Thesis:</i> Embodied language agents for real-world robotic tasks: visually-grounded interaction & skill learning	2020 – 2025
	Seoul National University M.S. in Cognitive Science <i>Advisor:</i> Byoung-Tak Zhang	2018 – 2020
	Ajou University B.Eng. in Software and Computer Engineering	2011 – 2018
INTERESTS	Machine Learning, Robot Learning, Embodied & Multimodal AI • Robot Foundation Models: Developing generalist robots by leveraging web-scale knowledge from foundation models • Human-Robot Interaction: Establishing natural, multimodal (verbal & non-verbal) interaction for shared autonomy • Lifelong Robot Learning: Continual skill learning and adaptation through interaction with the real world and humans	
FORMER POSITIONS	Postdoctoral Fellow , SNU AI Institute <i>Project:</i> Vision-language-action (VLA) models for robot manipulation	2025 – 2026
	Student Researcher , NAVER AI <i>Project:</i> Semi-supervised learning for visual dialog	Summer 2022
	Research Assistant , Seoul National University <i>Project:</i> Machine learning for multimodal and embodied AI	2020 – 2025
	Research Intern , SK T-Brain <i>Project:</i> Discovering sparse structures in visual dialog	Winter 2020
HONORS & AWARDS	RSS Pioneers <i>Top 33 early-career researchers in robotics worldwide (acceptance rate: 14%)</i>	2025
	Youlchon AI Star Fellowship <i>Awarded to 10 PhD students for excellence in core AI research</i>	2024
	IPAI Ph.D. Fellowship	2024
	Daeha Scholarship	2022
	3rd Place, Visual Dialog Challenge @ CVPR 2019	2019
	Software-Centered University Hackathon, Prize from Microsoft Korea	2017
	Yeoju Honor Scholarship	2016

SELECTED PUBLICATIONS

(* equal contribution, † co-advising)

PEER-REVIEWED CONFERENCE PAPERS

- [1] **Disentangling Spurious Correlations in Vision-Language-Action Models via Predicting Domain-Invariant Latent Lookahead**
Junghyun Kim*, Ngseo Kim*, Seoyeon Lee, ChungWoo Lee, Woo-Jeong Baek, Adam Zhou, Jun-Ki Lee†, Gi-Cheon Kang†, Byoung-Tak Zhang†
Conference on Robot Learning (CoRL), 2026
RSS Workshop on SemRob, 2026 **(Spotlight)**
- [2] **Continual Vision-and-Language Navigation**
Seongjun Jeong, Gi-Cheon Kang, Seongho Choi, Joochan Kim, Byoung-Tak Zhang
The British Machine Vision Conference (BMVC), 2025
- [3] **CLIP-RT: Learning Language-Conditioned Robotic Policies from Natural Language Supervision**
Gi-Cheon Kang*, Junghyun Kim*, Kyuhwan Shim, Jun Ki Lee, Byoung-Tak Zhang
Robotics: Science and Systems (RSS), 2025 **(Oral)**
- [4] **Socratic Planner: Self-QA-Based Zero-Shot Planning for Embodied Instruction Following**
Suyeon Shin, Sujin Jeon*, Junghyun Kim*, Gi-Cheon Kang*, Byoung-Tak Zhang
IEEE International Conference on Robotics and Automation (ICRA), 2025
- [5] **PGA: Personalizing Grasping Agents with Single Human-Robot Interaction**
Junghyun Kim, Gi-Cheon Kang*, Jaein Kim*, Seoyoon Yang, Minjoon Jung, Byoung-Tak Zhang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
- [6] **PROGrasp: Pragmatic Human-Robot Communication for Object Grasping**
Gi-Cheon Kang, Junghyun Kim, Jaein Kim, Byoung-Tak Zhang
IEEE International Conference on Robotics and Automation (ICRA), 2024 **(Oral)**
- [7] **The Dialog Must Go On: Improving Visual Dialog via Generative Self-Training**
Gi-Cheon Kang, Sungdong Kim*, Jin-Hwa Kim*, Donghyun Kwak*, Byoung-Tak Zhang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- [8] **GVCCI: Lifelong Learning of Visual Grounding for Language-Guided Robotic Manipulation**
Junghyun Kim, Gi-Cheon Kang*, Jaein Kim*, Suyeon Shin, Byoung-Tak Zhang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023 **(Oral)**
- [9] **Reasoning Visual Dialog with Sparse Graph Learning and Knowledge Transfer**
Gi-Cheon Kang, Junseok Park, Hwaran Lee, Byoung-Tak Zhang*, Jin-Hwa Kim*
Findings of Empirical Methods in Natural Language Processing (EMNLP Findings), 2021
- [10] **Attend What You Need: Motion-Appearance Synergistic Networks for Video Question Answering**
Ahjeong Seo, Gi-Cheon Kang, Joonhan Park, Byoung-Tak Zhang
Annual Meeting of the Association for Computational Linguistics (ACL), 2021

- [11] **Label Propagation Adaptive Resonance Theory for Semi-Supervised Continuous Learning**
Taehyeong Kim, Injune Hwang, Gi-Cheon Kang, Won-Seok Choi, Hyunseo Kim, Byoung-Tak Zhang
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020
- [12] **Dual Attention Networks for Visual Reference Resolution in Visual Dialog**
Gi-Cheon Kang, Jaeseo Lim, Byoung-Tak Zhang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2019
ICCV Workshop on Video Turing Test (VTT), 2019 **(Spotlight)**

WORKSHOP PAPERS

- [13] **MicroVLA: Edge-Deployable Vision Language Action at 10M Parameters**
Ngseo Kim, Junghyun Kim, Gi-Cheon Kang[†], Youngjae Yu[†], Byoung-Tak Zhang[†]
RSS Workshop on Lab to Production (RSSW), 2026
- [14] **Can Embodied Agents Remember What You Said? Evaluating Dialogue-Grounded Embodied Memory in 3D Environments**
Seongjun Jeong, Junghyun Kim, Gi-Cheon Kang[†], Byoung-Tak Zhang[†]
CVPR Workshop on 3D-LLM/VLA (CVPRW), 2026
- [15] **Language-Driven Robotics: Learning and Interaction**
Gi-Cheon Kang
RSS Pioneers Workshop (RSSW), 2025
- [16] **Generative Self-training Improves Pre-training for Visual Dialog**
Gi-Cheon Kang, Sungdong Kim*, Jin-Hwa Kim*, Donghyun Kwak*, Byoung-Tak Zhang
ICML Workshop on Pre-training: Perspectives, Pitfalls, and Paths Forward (ICMLW), 2022
- [17] **Improving Robustness to Texture Bias via Shape-focused Augmentation**
Sangjun Lee, Inwoo Hwang, Gi-Cheon Kang, Byoung-Tak Zhang
CVPR Workshop on Human-centered Intelligent Services: Safety and Trustworthy (CVPRW), 2022
- [18] **C³: Contrastive Learning for Cross-domain Correspondence in Few-shot Image Generation**
Hyukgi Lee, Gi-Cheon Kang, Chang-Hoon Jeong, Hanwool Sul, Byoung-Tak Zhang
NeurIPS Workshop on Controllable Generative Modeling in Language and Vision (NeurIPSW), 2021
- [19] **Contextualized Bilinear Attention Networks**
Gi-Cheon Kang, Seonil Son, Byoung-Tak Zhang
ECCV Workshop on VizWiz Grand Challenge (ECCVW), 2018

SERVICES	Reviewing	
	IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2026
	IEEE Transactions on Robotics (T-RO)	2025
	IEEE Robotics and Automation Letters (RA-L)	2024
	Conference on Robot Learning (CoRL)	2026
	Robotics: Science and Systems (RSS)	2026
	Conference on Neural Information Processing Systems (NeurIPS)	2023 – 2025
	International Conference on Machine Learning (ICML)	2024 – 2025
	International Conference on Learning Representations (ICLR)	2024 – 2026
	AAAI Conference on Artificial Intelligence (AAAI)	2025
	ACL Rolling Review (ARR)	2024 – 2025
INVITED TALKS	<i>Toward Human-Centered Vision-Language-Action Models</i>	
	Semiconductor Institute, Korea Institute of Science and Technology (KIST)	May 2026
	<i>Vision-Language-Action Models for General-Purpose Robot Manipulation</i>	
	Dept. of Immersive Media Engineering, Sungkyunkwan University	Nov. 2025
	Graduate School of AI, UNIST	Oct. 2025
	<i>CLIP-RT: Learning Language-Conditioned Robotic Policies from Natural Language Supervision</i>	
	Robotics: Science and Systems (RSS)	June 2025
	<i>Foundation Models for Robotics: Learning and Interaction</i>	
	Robotics: Science and Systems (RSS) Pioneers Workshop	June 2025
	<i>Vision-Language-Action Models for Robotic Manipulation</i>	
	Korea Robotics Society (KROS) Workshop on Mobile Manipulation	May 2025
EXTRA CURRICULAR ACTIVITIES	<i>PROGrasp: Pragmatic Human-Robot Communication for Object Grasping</i>	
	IEEE International Conference on Robotics and Automation (ICRA)	May 2024
	<i>The Evolution of Language Models: From Basic NLP to ChatGPT and Beyond</i>	
	Dept. of Energy Resources Engineering, Seoul National University	Nov. 2023
	<i>The Dialog Must Go On: Improving Visual Dialog via Generative Self-Training</i>	
	IEEE RO-MAN Workshop on Learning by Asking for Intelligent Robots and Agents	Aug. 2023
	<i>Reasoning Visual Dialog with Sparse Graph Learning and Knowledge Transfer</i>	
	KSC 2021—Top-tier Conference Paper Presentation Session	Dec. 2021
	Annual Conference on Human and Cognitive Language Technology	Oct. 2021
	<i>Dual Attention Networks for Visual Reference Resolution in Visual Dialog</i>	
	ICCV 2019 - Video Turing Test Workshop	Nov. 2019
	SK Telecom AI Center	Sept. 2019
	BI Lab Tennis Club	2023 – 2025
	Military Service in Republic of Korea Army	2012 – 2014